

总习题四

基础题

计算下列不定积分

$$\begin{aligned} 1. \int \frac{1}{1+\sqrt{x^2+2x+2}} dx &= \int \frac{1}{1+\sqrt{(x+1)^2+1}} dx \quad \underline{t=x+1} \quad \int \frac{\sec^2 t}{1+\sec t} dt \\ &= \int \frac{1}{(\cos t+1)\cos t} dt = \int \left(\frac{1}{\cos t} - \frac{1}{\cos t+1} \right) dt = \int \left(\sec t - \frac{1-\cos t}{\sin^2 t} \right) dt \\ &= \int (\sec t - \csc^2 t + \cot t \csc t) dt = \ln|\sec t + \tan t| + \cot t - \csc t + C \\ &= \ln \left| x+1+\sqrt{x^2+2x+2} \right| + \frac{1}{x+1} - \frac{\sqrt{x^2+2x+2}}{x+1} + C. \end{aligned}$$

$$\begin{aligned} 2. \int \frac{1}{1+\sqrt[3]{x+2}} dx \quad \underline{t=\sqrt[3]{x+2}} \quad 3 \int \frac{t^2}{1+t} dt &= 3 \int \left(t-1+\frac{1}{1+t} \right) dt = \frac{3}{2} t^2 - 3t + 3 \ln|1+t| + C \\ &= \frac{3}{2} \sqrt[3]{(x+2)^2} - 3\sqrt[3]{x+2} + 3 \ln|1+\sqrt[3]{x+2}| + C \end{aligned}$$

$$\begin{aligned} 3. \int \sqrt{\frac{x}{1-x\sqrt{x}}} dx \quad \underline{\sqrt{x}=t} \quad \int \frac{2}{\sqrt{1-t^3}} \cdot 2t dt &= -\frac{2}{3} \int \frac{d(1-t^3)}{\sqrt{1-t^3}} \\ &= -\frac{4}{3} \sqrt{1-t^3} + C = -\frac{4}{3} \sqrt{1-x\sqrt{x}} + C. \end{aligned}$$

$$\begin{aligned} 4. \int \frac{x^5}{\sqrt{1+x^2}} dx &= \int \frac{x^4 \cdot x}{\sqrt{1+x^2}} dx = \frac{1}{2} \int \frac{(1+x^2-1)^2}{\sqrt{1+x^2}} d(1+x^2) \\ &= \frac{1}{2} \int [(1+x^2)^{\frac{3}{2}} - 2\sqrt{1+x^2} + \frac{1}{\sqrt{1+x^2}}] d(1+x^2) \\ &= \frac{1}{5} (1+x^2)^{\frac{5}{2}} - \frac{2}{3} (1+x^2)^{\frac{3}{2}} + \sqrt{1+x^2} + C \end{aligned}$$

$$\begin{aligned} 5. \int \frac{1}{x^2 \sqrt{x^2+a^2}} dx \quad \underline{x=\frac{1}{t}} \quad - \int \frac{t}{\sqrt{a^2 t^2+1}} dt &= -\frac{1}{2a^2} \int \frac{d(a^2 t^2+1)}{\sqrt{a^2 t^2+1}} = -\frac{1}{a^2} \sqrt{a^2 t^2+1} + C \\ &= -\frac{\sqrt{x^2+a^2}}{a^2 x} + C \end{aligned}$$

$$6. \int \frac{1}{2x+\sqrt{1-x^2}} dx \quad \underline{x=\sin t} \quad \int \frac{\cos t}{2\sin t+\cos t} dt$$

$$\begin{aligned}
&= \frac{1}{5} \int \frac{(2 \sin t + \cos t) + 2(2 \cos t - \sin t)}{2 \sin t + \cos t} dt \\
&= \frac{1}{5} \int dt + \frac{1}{5} \int \frac{d(2 \sin t + \cos t)}{2 \sin t + \cos t} = \frac{1}{5} [t + \ln |2 \sin t + \cos t|] + C \\
&= \frac{1}{5} [\arcsin x + \ln |2x + \sqrt{1-x^2}|] + C
\end{aligned}$$

$$\begin{aligned}
7. \quad \int \frac{1}{(2x^2+1)(\sqrt{1+x^2})} dx &\stackrel{x=\tan t}{=} \int \frac{\sec^2 t}{(2 \tan^2 t + 1) \sec t} dt = \int \frac{\cos t}{2 \sin^2 t + \cos^2 t} dt = \int \frac{\cos t}{1 + \sin^2 t} dt \\
&= \int \frac{d \sin t}{1 + \sin^2 t} = \arctan \sin t + C = \arctan \frac{x}{\sqrt{1+x^2}} + C
\end{aligned}$$

$$\begin{aligned}
8. \quad \int \frac{1}{\sqrt{x}} \sqrt[3]{1+\sqrt[4]{x}} dx &\stackrel{t=\sqrt[4]{x}}{=} \int \frac{1}{t^2} \cdot \sqrt[3]{1+t} \cdot 4t^3 dt = 4 \int t \sqrt[3]{1+t} dt = 4 \int (t+1-1) \sqrt[3]{1+t} dt \\
&= 4 \int [(t+1)^{\frac{4}{3}} - (t+1)^{\frac{1}{3}}] dt = \frac{12}{7} (t+1)^{\frac{7}{3}} - 3(t+1)^{\frac{4}{3}} + C \\
&= \frac{12}{7} (\sqrt[4]{x}+1)^{\frac{7}{3}} - 3(\sqrt[4]{x}+1)^{\frac{4}{3}} + C
\end{aligned}$$

$$\begin{aligned}
9. \quad \int x^2 \arctan x dx &= \int \arctan x d\left(\frac{x^3}{3}\right) = \frac{x^3}{3} \arctan x - \frac{1}{3} \int \frac{x^3}{1+x^2} dx \\
&= \frac{x^3}{3} \arctan x - \frac{1}{6} \int \frac{(1+x^2)-1}{1+x^2} d(1+x^2) = \frac{x^3}{3} \arctan x - \frac{x^2}{6} + \frac{\ln(1+x^2)}{6} + C
\end{aligned}$$

$$\begin{aligned}
10. \quad \int \frac{\ln^3 x}{x^2} dx &= - \int \ln^3 x d\left(\frac{1}{x}\right) = -\frac{\ln^3 x}{x} + 3 \int \frac{\ln^2 x}{x^2} dx = -\frac{\ln^3 x}{x} - 3 \int \ln^2 x d\left(\frac{1}{x}\right) \\
&= -\frac{\ln^3 x}{x} - \frac{3 \ln^2 x}{x} - 6 \int \ln x d\left(\frac{1}{x}\right) = -\frac{\ln^3 x}{x} - \frac{3 \ln^2 x}{x} - \frac{6 \ln x}{x} + 6 \int \frac{1}{x^2} dx \\
&= -\frac{1}{x} (\ln^3 x + 3 \ln^2 x + 6 \ln x + 6) + C
\end{aligned}$$

$$\begin{aligned}
11. \quad \int \frac{\ln \cos x}{\cos^2 x} dx &= \int \ln \cos x \tan x = \tan x \cdot \ln \cos x + \int \tan^2 x dx \\
&= \tan x \cdot \ln \cos x + \int (\sec^2 x - 1) dx = \tan x \cdot \ln \cos x + \tan x - x + C
\end{aligned}$$

$$\begin{aligned}
12. \quad \int x^3 (2x+1)^9 dx &= \frac{1}{16} \int (2x+1-1)^3 (2x+1)^9 d(2x+1) \\
&= \frac{1}{16} \int [(2x+1)^3 - 3(2x+1)^2 + 3(2x+1) - 1] (2x+1)^9 d(2x+1) \\
&= \frac{1}{16} \int [(2x+1)^{12} - 3(2x+1)^{11} + 3(2x+1)^{10} - (2x+1)^9] d(2x+1) \\
&= \frac{(2x+1)^{13}}{208} - \frac{(2x+1)^{12}}{64} + \frac{3}{176} (2x+1)^{11} - \frac{(2x+1)^{10}}{160} + C
\end{aligned}$$

$$\begin{aligned}
 13. \quad \int \frac{x^4}{(x+2)^3} dx & \quad \underline{x+2=t} \quad \int \frac{(t-2)^4}{t^3} dt = \int \frac{(t-2)^4}{t^3} dt = \int (t-8+\frac{24}{t}-\frac{32}{t^2}+\frac{16}{t^3}) dt \\
 & = \frac{t^2}{2} - 8t + 24 \ln|t| + \frac{32}{t} - \frac{8}{t^2} + C = \frac{(x+2)^2}{2} - 8(x+2) + 24 \ln|x+2| + \frac{32}{x+2} - \frac{8}{(x+2)^2} + C
 \end{aligned}$$

$$\begin{aligned}
 14. \quad \int \frac{\operatorname{arccot} e^x}{e^x} dx & = -\int \operatorname{arccot} e^x de^{-x} = -e^{-x} \cdot \operatorname{arccot} e^x - \int \frac{1}{1+e^{2x}} dx \\
 & = -e^{-x} \cdot \operatorname{arccot} e^x - \int \frac{1+e^{2x}-e^{2x}}{1+e^{2x}} dx = -e^{-x} \cdot \operatorname{arccot} e^x - x + \frac{1}{2} \ln(1+e^{2x}) + C
 \end{aligned}$$

$$\begin{aligned}
 15. \quad \int \frac{2x^3+2x^2+5x+5}{x^4+5x^2+4} dx & = \int (\frac{x+1}{x^2+1} + \frac{x+1}{x^2+4}) dx \\
 & = \frac{1}{2} \ln(x^4+5x^2+4) + \frac{1}{2} \arctan \frac{x}{2} + \arctan x + C
 \end{aligned}$$

$$\begin{aligned}
 16. \quad \int \frac{x^3+1}{x(x-1)^3} dx & = \int (\frac{-1}{x} + \frac{2}{x-1} + \frac{1}{(x-1)^2} + \frac{2}{(x-1)^3}) dx \\
 & = -\ln|x| + 2 \ln|x-1| - \frac{1}{x-1} - \frac{1}{(x-1)^2} + C
 \end{aligned}$$

$$\begin{aligned}
 17. \quad \int \frac{x^3-2x^2+4x-1}{x^2-2x+5} dx & = \int \frac{x(x^2-2x+5)-x-1}{x^2-2x+5} dx = \int (x - \frac{1}{2} \cdot \frac{(2x-2)+4}{x^2-2x+5}) dx \\
 & = \frac{x^2}{2} - \frac{1}{2} \int \frac{d(x^2-2x+5)}{x^2-2x+5} - 2 \int \frac{dx}{x^2-2x+5} = \frac{x^2}{2} - \frac{1}{2} \ln(x^2-2x+5) - 2 \int \frac{d(x-1)}{(x-1)^2+2^2} \\
 & = \frac{x^2}{2} - \frac{1}{2} \ln(x^2-2x+5) - \arctan \frac{x-1}{2} + C
 \end{aligned}$$

$$\begin{aligned}
 18. \quad \int \frac{x^2}{(x-1)^{10}} dx & \quad \underline{x-1=t} \quad \int \frac{(t+1)^2}{t^{10}} dt = \int \frac{t^2+2t+1}{t^{10}} dt = \int (t^{-8}+2t^{-9}+t^{-10}) dt \\
 & = -\frac{1}{7} t^{-7} - \frac{1}{4} t^{-8} - \frac{1}{9} t^{-9} + C = -\frac{1}{7} (x-1)^{-7} - \frac{1}{4} (x-1)^{-8} - \frac{1}{9} (x-1)^{-9} + C
 \end{aligned}$$

$$\begin{aligned}
 19. \quad \int \frac{x^{11}}{x^8+3x^4+2} dx & = \int \frac{x^8 \cdot x^3}{x^8+3x^4+2} dx = \frac{1}{4} \int \frac{x^8}{x^8+3x^4+2} dx \\
 & = \frac{1}{4} \int (1 - \frac{1}{x^4+1} + \frac{4}{x^4+2}) dx = \frac{1}{4} [x^4 - \ln(x^4+1) + 4 \ln(x^4+2)] + C
 \end{aligned}$$

$$\begin{aligned}
20. \quad \int \frac{x^2 + 3x - 2}{(x-1)(x^2 + x + 1)^2} dx &= \frac{5x+2}{3(x^2 + x + 1)} + \frac{2}{9} \int \frac{dx}{x-1} - \frac{1}{9} \int \frac{2x-11}{x^2 + x + 1} dx \\
&= \frac{5x+2}{3(x^2 + x + 1)} + \frac{2}{9} \ln|x-1| - \frac{1}{9} \int \frac{2x+1}{x^2 + x + 1} dx + \frac{4}{3} \int \frac{dx}{(x+\frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2} \\
&= \frac{5x+2}{3(x^2 + x + 1)} + \frac{2}{9} \ln|x-1| - \frac{1}{9} \ln|x^2 + x + 1| + \frac{8}{3\sqrt{3}} \arctan \frac{2x+1}{\sqrt{3}} + C
\end{aligned}$$

$$\begin{aligned}
21. \quad \int \frac{dx}{2\sin x - \cos x} \quad \underline{t = \tan \frac{x}{2}} \quad \int \frac{1}{2 \cdot \frac{2t}{1+t^2} - \frac{1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2} dt = \int \frac{2}{t^2 + 4t - 1} dt \\
= \int \frac{2}{(t+2)^2 - (\sqrt{5})^2} d(t+2) = \frac{1}{\sqrt{5}} \ln \left| \frac{t+2-\sqrt{5}}{t+2+\sqrt{5}} \right| + C = \frac{1}{\sqrt{5}} \ln \left| \frac{\tan \frac{x}{2} + 2 - \sqrt{5}}{\tan \frac{x}{2} + 2 + \sqrt{5}} \right| + C
\end{aligned}$$

$$\begin{aligned}
22. \quad \int \frac{dx}{\sin 2x + 2\sin x} \quad \underline{t = \tan \frac{x}{2}} \quad \int \frac{1}{2 \cdot \frac{2t}{1+t^2} \cdot \frac{1-t^2}{1+t^2} + 2 \cdot \frac{2t}{1+t^2}} \cdot \frac{2}{1+t^2} dt \\
= \frac{1}{4} \int \left(\frac{1}{t} + t \right) dt = \frac{1}{4} (\ln|t| + \frac{1}{2} t^2) + C = \frac{1}{4} \ln \left| \tan \frac{x}{2} \right| + \frac{1}{8} \tan^2 \frac{x}{2} + C
\end{aligned}$$

$$\begin{aligned}
23. \quad \int \frac{dx}{(1+\sqrt[3]{x})\sqrt{x}} \quad \underline{x=t^6} \quad \int \frac{6t^5}{(1+t^2)t^3} dt = 6 \int \frac{t^2}{1+t^2} dt \\
= 6(t - \arctan t) + C = 6(\sqrt[6]{x} - \arctan \sqrt[6]{x}) + C
\end{aligned}$$

$$\begin{aligned}
24. \quad \int \frac{dx}{2\cos^2 x + \sin x \cos x + \sin^2 x} = \int \frac{dx}{\cos^2 x (2 + \tan x + \tan^2 x)} = \int \frac{d\tan x}{2 + \tan x + \tan^2 x} \\
= \int \frac{d\tan x}{(\tan x + \frac{1}{2})^2 + (\frac{\sqrt{7}}{2})^2} = \frac{2}{\sqrt{7}} \arctan \frac{2\tan x + 1}{\sqrt{7}} + C
\end{aligned}$$

$$\begin{aligned}
25. \quad \int e^{2x} \tan x (2 + \tan x) dx &= 2 \int e^{2x} \tan x dx + \int e^{2x} \tan^2 x dx = \int \tan x de^{2x} + \int e^{2x} \tan^2 x dx \\
&= e^{2x} \tan x - \int e^{2x} \sec^2 x dx + \int e^{2x} \tan^2 x dx = e^{2x} \tan x - \int e^{2x} \tan^2 x dx - \int e^{2x} dx + \int e^{2x} \tan^2 x dx \\
&= e^{2x} \tan x - \frac{1}{2} e^{2x} + C
\end{aligned}$$

$$\begin{aligned}
26. \quad \int \frac{dx}{\sin^3 x \cdot \cos^5 x} &= \int \frac{\sin^2 x + \cos^2 x}{\sin^3 x \cdot \cos^5 x} dx = \int \frac{dx}{\sin x \cdot \cos^5 x} + \int \frac{dx}{\sin^3 x \cdot \cos^3 x} \\
&= \int \frac{\sin^2 x + \cos^2 x}{\sin x \cdot \cos^5 x} dx + \int \frac{\sin^2 x + \cos^2 x}{\sin^3 x \cdot \cos^3 x} dx \\
&= \int \left(\frac{2 \sin x}{\cos^3 x} + \frac{\sin x}{\cos^5 x} + \frac{\cos x}{\sin^3 x} + \frac{3 \sin x}{\cos x} + \frac{3 \cos x}{\sin x} \right) dx \\
&= \frac{1}{\cos^2 x} + \frac{1}{4 \cos^4 x} - \frac{1}{2 \sin^2 x} - 3 \ln |\cos x| + 3 \ln |\sin x| + C
\end{aligned}$$

$$\begin{aligned}
27. \quad \int \frac{dx}{1 + \sqrt{x} + \sqrt{x+1}} \quad t = \sqrt{x} + \sqrt{x+1} \quad \frac{1}{2} \int \frac{t^4 - 1}{(1+t)t^3} dt &= \frac{1}{2} \int \left(1 - \frac{1}{t} + \frac{1}{t^2} - \frac{1}{t^3} \right) dt \\
&= \frac{1}{2} \left(t - \ln t - \frac{1}{t} + \frac{1}{2t^2} \right) + C_1 = \sqrt{x} - \frac{1}{2} \ln(\sqrt{x} + \sqrt{x+1}) + \frac{x}{2} - \frac{1}{2} \sqrt{x(x+1)} + C
\end{aligned}$$

28.

$$\begin{aligned}
&\int \ln \left(1 + \sqrt{\frac{1+x}{x}} \right) dx \quad \sqrt{\frac{1+x}{x}} = t \quad = \int \ln(1+t) d \frac{1}{t^2-1} = \frac{\ln(1+t)}{t^2-1} - \int \frac{1}{(t^2-1)(t+1)} dt \\
&= \frac{\ln(1+t)}{t^2-1} - \frac{1}{4} \int \left[\frac{1}{t-1} - \frac{1}{t+1} - \frac{2}{(t+1)^2} \right] dt = \frac{\ln(1+t)}{t^2-1} - \frac{1}{4} \ln|t-1| + \frac{1}{4} \ln|t+1| - \frac{1}{2(t+1)} + C \\
&= x \ln \left(1 + \sqrt{\frac{1+x}{x}} \right) + \frac{1}{2} \ln(\sqrt{1+x} + \sqrt{x}) + \frac{1}{2} x - \frac{1}{2} \sqrt{x+x^2} + C
\end{aligned}$$

$$\begin{aligned}
29. \quad \int e^x \left(\frac{1-x}{1+x^2} \right)^2 dx &= \int e^x \frac{1-2x+x^2}{(1+x^2)^2} dx = \int \frac{e^x(1+x^2) - 2xe^x}{(1+x^2)^2} dx \\
&= \int \left(\frac{e^x}{1+x^2} \right)' dx = \frac{e^x}{1+x^2} + C
\end{aligned}$$

$$\begin{aligned}
30. \quad \int \frac{\arctan e^{\frac{x}{2}}}{e^{\frac{x}{2}}(1+e^x)} dx &= \int \left(e^{\frac{x}{2}} - \frac{e^{\frac{x}{2}}}{(1+e^x)} \right) \arctan e^{\frac{x}{2}} dx \\
&= -2 \int \arctan e^{\frac{x}{2}} de^{\frac{x}{2}} - 2 \int \arctan e^{\frac{x}{2}} d \arctan e^{\frac{x}{2}} = -2e^{\frac{x}{2}} \arctan e^{\frac{x}{2}} + \int \frac{dx}{1+e^x} - (\arctan e^{\frac{x}{2}})^2 \\
&= -2e^{\frac{x}{2}} \arctan e^{\frac{x}{2}} + x - \ln(1+e^x) - (\arctan e^{\frac{x}{2}})^2 + C
\end{aligned}$$

提高题

$$\begin{aligned}
 1. \quad I_n &= \int \sec^n x dx = \int \sec^{n-2} x \cdot \sec^2 x dx = \int \sec^{n-2} x d \tan x \\
 &= \sec^{n-2} x \cdot \tan x - (n-2) \int \sec^{n-3} x \cdot \sec x \tan x \cdot \tan x dx \\
 &= \sec^{n-2} x \cdot \tan x - (n-2) \int \sec^{n-2} x (\sec^2 x - 1) dx \\
 &= \sec^{n-2} x \cdot \tan x - (n-2) I_n + (n-2) I_{n-2}
 \end{aligned}$$

$$\text{所以} \quad I_n = \frac{1}{n-1} \sec^{n-2} x \cdot \tan x + \frac{n-2}{n-1} I_{n-2} \quad (n \geq 2).$$

2.

$$\begin{aligned}
 \int x^3 f'(x) dx &= \int x^3 df(x) = x^3 f(x) - 3 \int x^2 f(x) dx = x^3 f(x) - 3 \int x^2 d\left(\frac{\sin x}{x}\right) \\
 &= x^3 f(x) - 3x^2 \cdot \frac{\sin x}{x} + 6 \int \sin x dx = x^3 \cdot \frac{x \cos x - \sin x}{x^2} - 3x \sin x - 6 \cos x + C \\
 &= x^2 \cos x - 4x \sin x - 6 \cos x + C.
 \end{aligned}$$